

of data and evidence, however, a common mistake that is made is to assume that evolution somehow proves that no creation ever occurred. This is incorrect. They're two separate topics. Take the example of the science fiction movie *Prometheus* (Spoiler alert if you haven't seen it). The movie starts with an Engineer standing on a barren (just rock and water) planet sacrificing himself by drinking a weird liquid that breaks his body and DNA down, and then the DNA reforms again with new connections, and life forms are born. It's a creation story that has evolution also thrown in. There are no supernatural claims, or claims of deities, just technology being used to kickstart life.

Is that where we are with AI currently? Are the primitive AIs that we have currently akin to us creating basic AI DNA, which will eventually evolve?

Evolution

If you're reading the cover stories linearly, you've already read Booman's take, and have heard him whinge about brute forcing, which is more popularly known as the infinite monkey theorem – basically the idea that random chance, if given infinite time, can create anything. Currently, AI software have been developed to experiment and modify their own codes so as to painstakingly check every possible modification and work out if it is better or not. Now, "better" is really a value judgement, so it's obvious that we have to set a goal in mind. A common example is image recognition, where the goal would be to get better at accurately matching images, or to get better at face recognition for example.

Assuming it's for facial recognition, in the beginning the AI has human-made code, and then it has to modify it slightly and check for accuracy. If the accuracy is higher, then it knows it is on the path to improvement. If it's worse, it knows it went in the wrong direction. Now, of course this isn't "intelligence",

it's brute forcing, just as Booman said it was – I completely agree with that. What I think we're all missing, however, is that this is an exact description of evolution and natural selection, which is how we evolved!

Think about it. Biological evolution always starts with a creature (or rather population) that has a set of genes, and then a slight mutation of those genes in its offspring cause the next generation to be slightly different. Then the offspring is subjected to natural selection, which, if it's better, favour its reproduction, and if it is worse, hinder that. Of course this doesn't work on the level of individuals, but more on large populations over very large periods of time, but that's just the same implementation or same scenario repeated several times. Exactly how the AI is doing it.

What's important to note with AI, however, is that the selection processes are artificial, and timelines are really, really fast...

Artificial Selection

There are some major differences between biological evolution and AI evolution. For starters, biological evolution has no real goal in mind except to survive and propagate genes, and has no guiding hand. Whereas for artificial selection, we are that guiding hand, and we're constantly meddling to change things. You would think this would speed AI up, but it very well might be hurting it. Imagine if aliens had a hand in evolution on earth. They might be more water-biased aliens, and prefer sea creatures because they found them cuter. Maybe they like shark videos the way we like cat videos, who knows? Would they cut off human evolution the minute we started killing and eating fish? The way we'd cut off an AI that displayed signs of racism, for instance – this actually happened with a Microsoft chatbot...

Of course, there's a flipside to this, because artificial selection can in fact be a good thing as well. It means a much faster pace of muta-

tion. It could also mean that there will be less time wasted exploring dead ends; something akin to the Dodo of AI evolution – unable to adapt to fast changes in environment, and thus unable to survive.

Then of course, there's a totally different aspect that most don't like to consider... is AI merely the next step in human evolution? Many futurists scoff at the idea that AI will replace humans. Instead, they believe that AI-empowered humans will replace humans! The idea is that we're headed towards a future where AI is merely a tool. Think of it as how we developed an opposable thumb, which distinguished us from our ape-like ancestors, and gave us an advantage. Similarly, AI is something that will empower humans, and give us advantages over others, and this will eventually lead to the extinction of humans as we know them.

Elon Musk is known to be fearful of a sentient general AI, but even he agrees that biological and digital intelligence will merge, in order to increase bandwidth. "It's mostly about the bandwidth, the speed of the connection between your brain and the digital version of yourself, particularly output", he told an audience who had gathered to witness a launch of his Tesla cars in Dubai, back in February 2017.

Whatever final path it takes, even if AI gains sentience and wipes out humanity, it seems obvious to me that AI will evolve and do it gradually. Heck, we may not even realise we're being replaced. Look at how we're applauding the entry of driverless cars... despite the fact that hundreds of millions of humans could essentially be jobless in a mere 20 to 40 years. Elon Musk may fear AI, but hundreds of millions of people across the world should fear his company. That's another reason I think this is more like evolution than anything else, because one feature of evolution is that it is cold, and unstoppable, and our best bet is to adapt, or perish. 